Polygenic Risk Scores

Haoyu Zhang

Earl Stadtman Tenure Track Investigator

Division of Cancer Epidemiology and Genetics

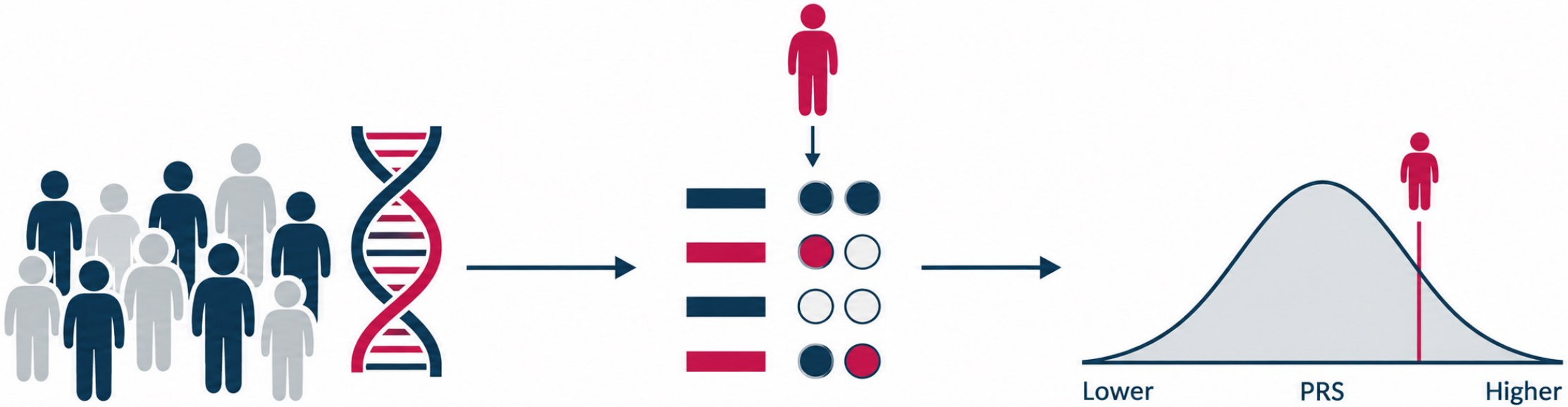
National Cancer Institute

NCI DCEG Statistical Genetics Workshops

NIH NATIONAL CANCER INSTITUTE

October 28, 2026

Polygenic scores summarize genetic effects



Score = sum of (allele count × estimated effect)

What does a high PRS mean?

90th percentile

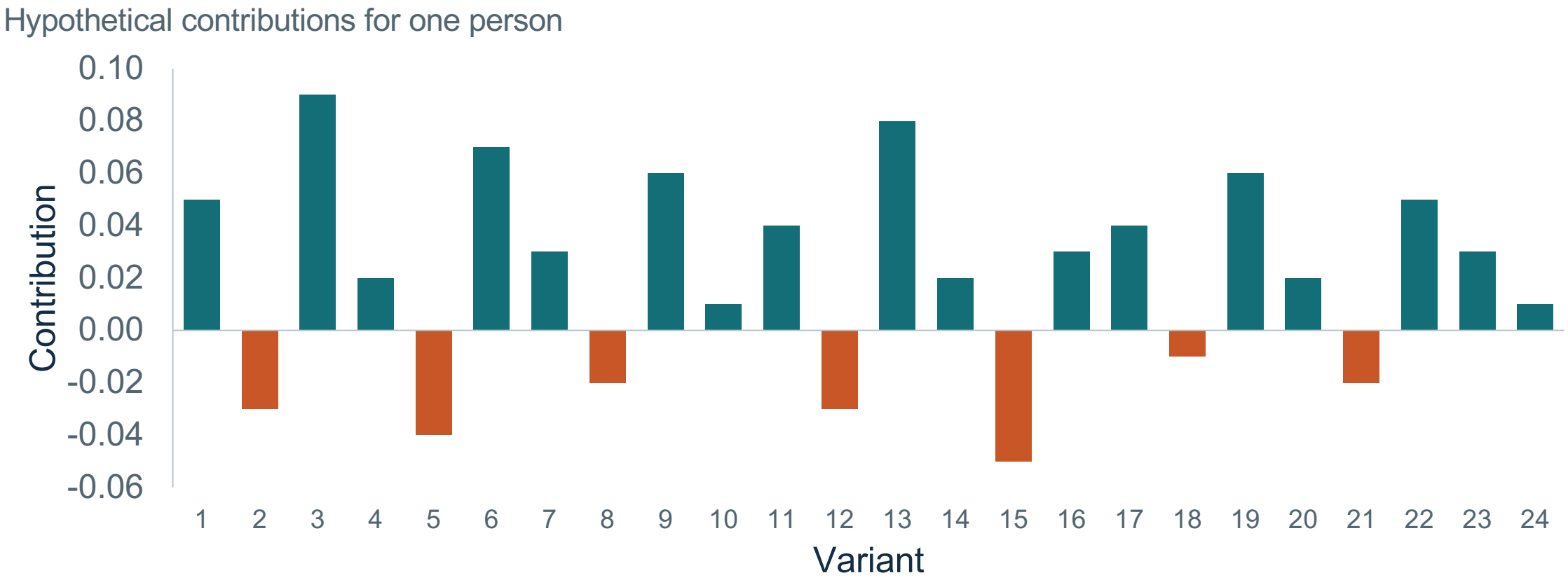
Higher score than 90% of a reference group

What else would you need to estimate disease risk?

Outline

- What a PRS measures
- Single-ancestry methods
- Evaluation and interpretation
- Multi-ancestry methods
- Continuous ancestry and rare variants

Many small contributions form a score



A person's PRS adds their weighted allele dosages.

Ge et al., Nat Commun, 2019; Wang et al., Nat Commun, 2020.

A PRS is a weighted sum of allele dosages

$$\mathrm{PRS}_i = \sum_{j=1}^{M} \widehat{\beta}_j G_{ij}$$

i	One person
$j = 1, \ldots, M$	Variants included in the score
G_{ij}	Effect-allele dosage, usually 0 to 2
$\widehat{\beta}_j$	Estimated weight for the effect allele

The effect allele can have a positive or a negative weight.

Choi et al., Nat Protoc, 2020.

A three-variant example

Variant	Effect allele	Dosage	Weight	Contribution
1	A	2	0.10	0.20
2	C	1	-0.20	-0.20
3	G	1	0.30	0.30

PRS = 0.20 - 0.20 + 0.30 = 0.30

Hypothetical example. A score of 0.30 does not mean 30% risk.

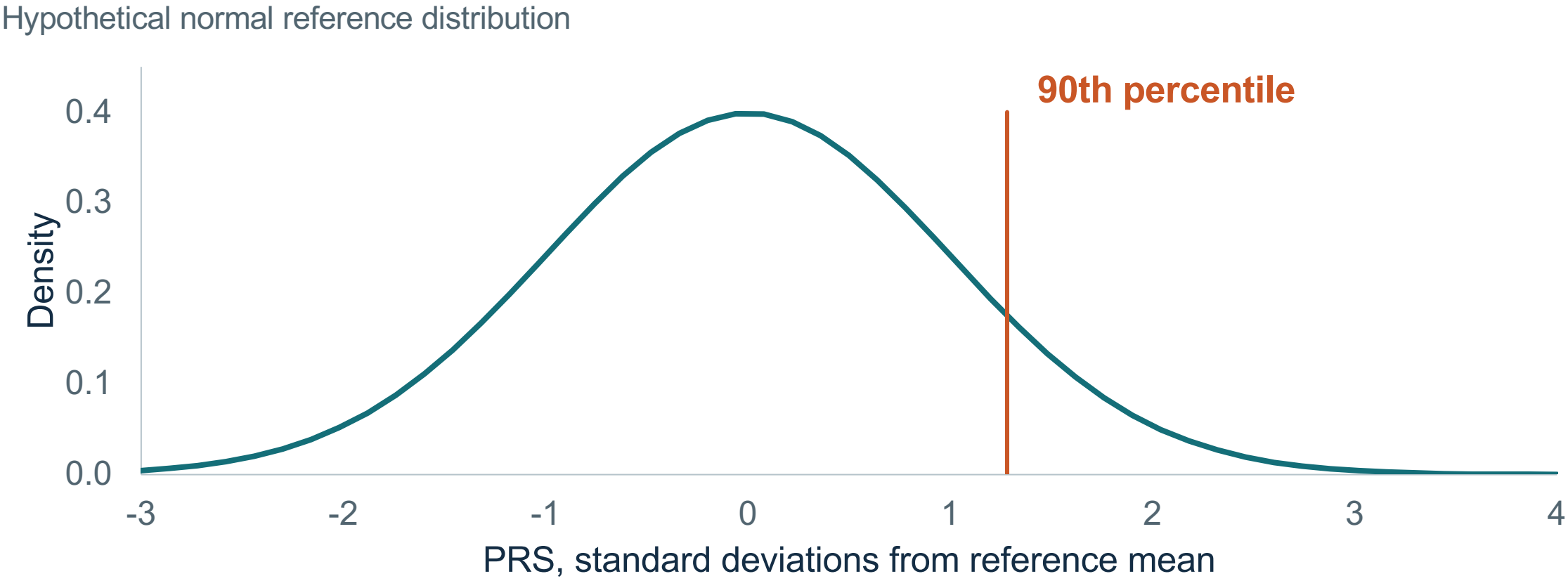
Activity: calculate a second person's score

Variant	Dosage	Weight
1	0	0.10
2	2	-0.20
3	1	0.30

PRS = 0.00 - 0.40 + 0.30 = -0.10

Calculate the sum. What can you conclude about this person's disease risk?

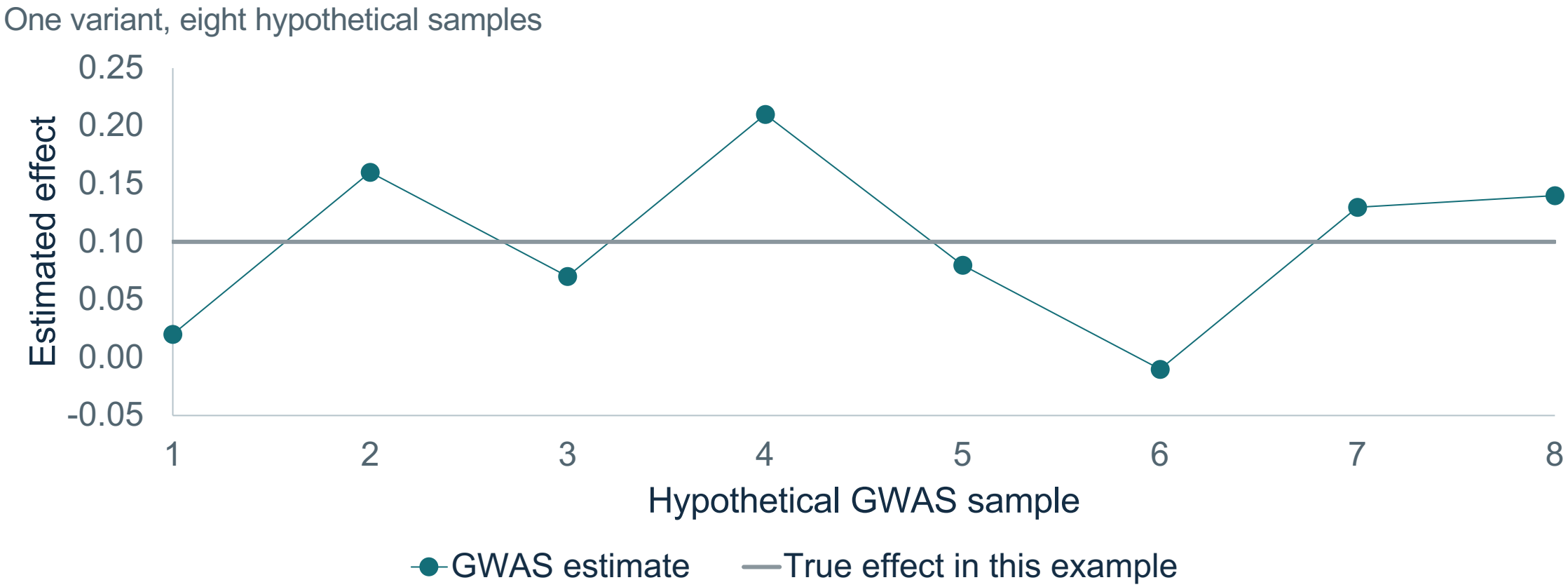
A percentile locates a score in a reference group



90th percentile: about 90% below this score, 10% above it.

Wand et al., Nature, 2021.

GWAS estimates contain sampling noise



PRS methods address noisy estimates and correlated variants.

Ge et al., Nat Commun, 2019; Privé et al., Bioinformatics, 2020.

Four data inputs

Input	Role
Discovery GWAS	Estimate associations
LD reference	Describe variant correlations
Tuning sample	Choose model settings
Independent validation sample	Evaluate the frozen model

Specify the target outcome and population first.

Choi et al., Nat Protoc, 2020; Wand et al., Nature, 2021.

Discovery, tuning and validation

Discovery	Estimate candidate SNP weights
Tuning	Choose settings and fit score combinations
Validation	Evaluate the frozen prediction model

Selecting a model using validation outcomes makes them development data.

Wand et al., Nature, 2021.

The weight must match the counted allele

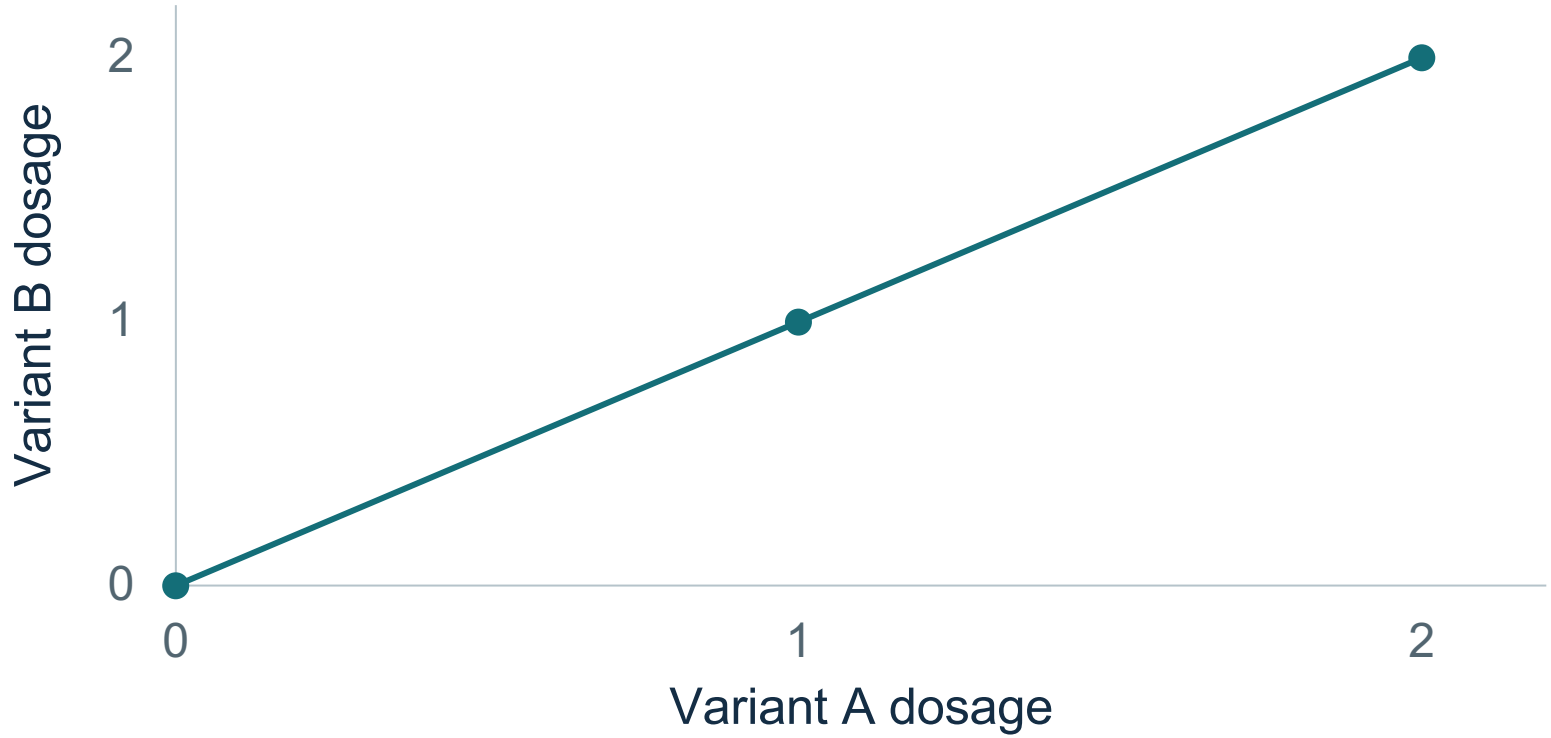
Score definition	Person’s genotype	Contribution
A allele, weight +0.20	A/G: one A	+0.20
Same definition	G/G: zero A	0.00

Counting G instead of A assigns the weight to the wrong allele.

Lambert et al., Nat Genet, 2021; Choi et al., Nat Protoc, 2020.

Correlated variants can repeat the same information

Perfect LD: A and B have identical dosages



With identical dosages, adding both marginal weights can count the signal twice.

The model behind summary-statistic PRS

Standardized quantitative trait y and genotypes X (N people × M variants)

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \qquad \mathbf{b} = \frac{\mathbf{X}^\top \mathbf{y}}{N} = \underbrace{\frac{\mathbf{X}^\top \mathbf{X}}{N}}_{\mathbf{R}} \boldsymbol{\beta} + \boldsymbol{\eta}$$

b: marginal GWAS effects. β: joint SNP effects.

R: LD correlation matrix. η: sampling noise.

Two variants with correlation 0.8

$$\mathrm{E}[\mathbf{b} \mid \mathbf{X}] = \mathbf{R}\boldsymbol{\beta} = \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix} \begin{pmatrix} 0.20 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.20 \\ 0.16 \end{pmatrix}$$

Variant 2 has a marginal association because it tags variant 1.

Ge et al., Nat Commun, 2019; Privé et al., Bioinformatics, 2020.

Clumping and thresholding: five variants

Variant	LD group	GWAS p-value	Final score
A	1	0.00000001	**Keep**
B	1	0.000001	Remove: tags A
C	2	0.001	**Keep**
D	2	0.04	Remove: tags C
E	3	0.20	Remove: p > 0.01

Clump correlated variants first. Apply the p-value cutoff next.

Choi et al., Nat Protoc, 2020.

Clumping and thresholding as a weight rule

$$\widehat{w}_j(t) = \widehat{\gamma}_j \, \mathbf{1}\{j \in C\} \, \mathbf{1}\{p_j \leq t\}$$

C: clumped variants. t: p-value cutoff selected in tuning data.

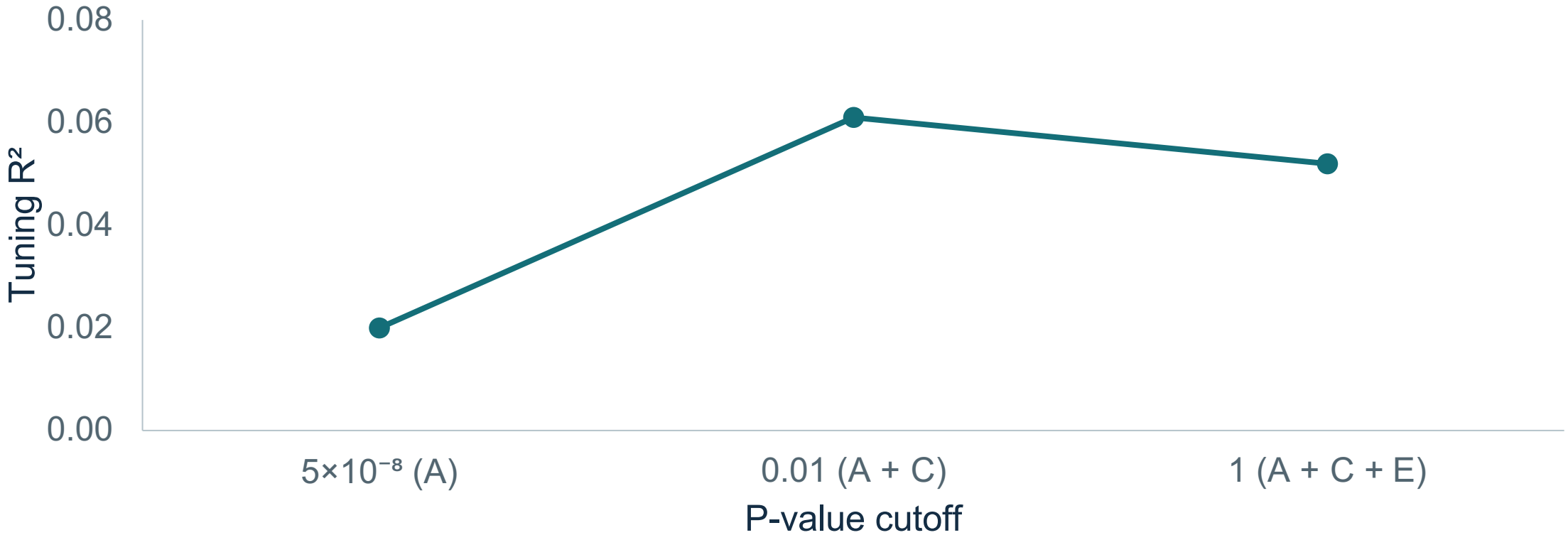
The aligned per-allele GWAS coefficient weights the raw allele dosage.

C+T selects weights. It does not jointly re-estimate correlated effects.

Choi et al., Nat Protoc, 2020.

The p-value cutoff is chosen in tuning data

Three retained-variant sets. R^2 is the fraction of outcome variation explained.



Hypothetical curve. Here, p ≤ 0.01 gives the highest tuning R^2.

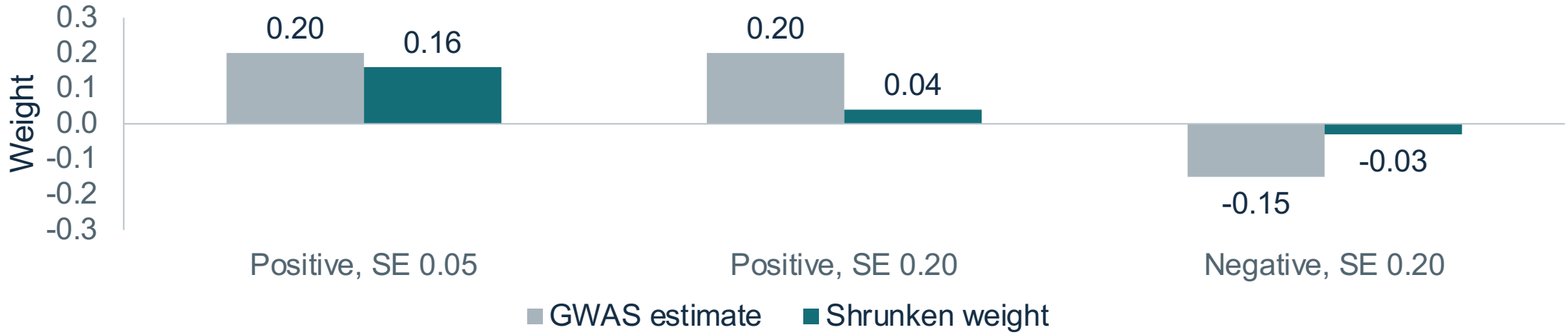
Shrinkage of SNP effect estimates

Shrinkage moves effect estimates toward zero.

$$\mathrm{E}[\beta \mid \widehat{\beta}] = \frac{\tau^2}{\tau^2 + s^2}\widehat{\beta}$$

One-SNP illustration: normal sampling error and a normal prior centered at zero.

β : underlying effect $\widehat{\beta}$: GWAS estimate s : standard error $\tau = 0.10$: prior SD



Holding the prior fixed, a larger standard error gives more shrinkage.

Ge et al., Nat Commun, 2019.

Likelihood, prior and posterior

Standardized trait and genotypes, independent Gaussian residuals, one sample

$$\mathbf{b} \mid \boldsymbol{\beta}, \mathbf{R}, \sigma^2 \sim \mathcal{N}\left(\mathbf{R}\boldsymbol{\beta}, \frac{\sigma^2}{N}\mathbf{R}\right)$$

b: marginal GWAS estimates. β: joint SNP effects. R: LD matrix.

N: discovery sample size. σ^2: residual variance.

LD affects both expected associations and their sampling uncertainty.

$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior}$$

LDpred2 and PRS-CS use posterior means, converted to the scoring scale.

Ge et al., Nat Commun, 2019; Privé et al., Bioinformatics, 2020.

LDpred2: a mixture prior on SNP effects

$$\beta_j \sim \underbrace{(1-p)\delta_0}_{\text{zero effect}} + \underbrace{p\,\mathcal{N}\left(0, \frac{h^2}{Mp}\right)}_{\text{nonzero effect}}$$

p: proportion of nonzero effects. h^2: SNP heritability. M: SNP count.

The LD model and this prior produce posterior mean weights.

Smaller p concentrates the same prior genetic variance in fewer SNPs.

Privé et al., Bioinformatics, 2020.

PRS-CS: global and SNP-specific shrinkage

$$\beta_j \mid \phi, \psi_j, \sigma^2 \sim \mathcal{N}\left(0, \frac{\sigma^2}{N} \underbrace{\phi}_{\text{global}} \underbrace{\psi_j}_{\text{local}}\right)$$

φ is shared across SNPs. The local scale ψ varies by SNP.

Holding other quantities fixed, a larger local scale gives less shrinkage.

Illustration: no LD, fixed scales, GWAS estimate = 0.20

$$\mathrm{E}[\beta_j \mid \widehat{\beta}_j, \phi, \psi_j] = \frac{\phi\psi_j}{1 + \phi\psi_j}\widehat{\beta}_j$$

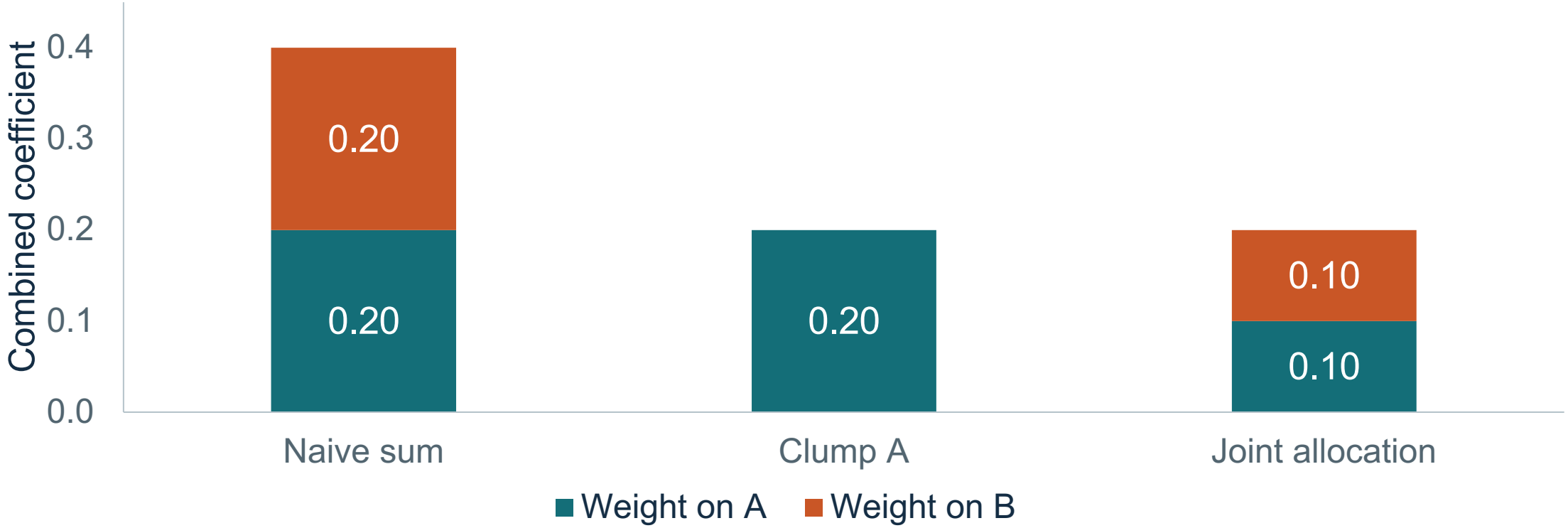
Combined scale	Multiplier	Shrunken estimate
0.25	0.20	0.04
4	0.80	0.16

Local scales are inferred jointly with effects using GWAS evidence and LD.

Ge et al., Nat Commun, 2019.

Selection and joint modeling handle LD differently

A and B have identical dosages. Both marginal weights are 0.20.



Perfect-LD teaching example. Joint weights are not uniquely identified without constraints.

Ge et al., Nat Commun, 2019; Privé et al., Bioinformatics, 2020.

The comparison is clinical model versus clinical model + PRS

Clinical predictors

Clinical predictors + PRS

How much does PRS add?

Use the same outcome, prediction time, population and evaluation sample.

Wand et al., Nature, 2021; Mavaddat et al., Am J Hum Genet, 2019.

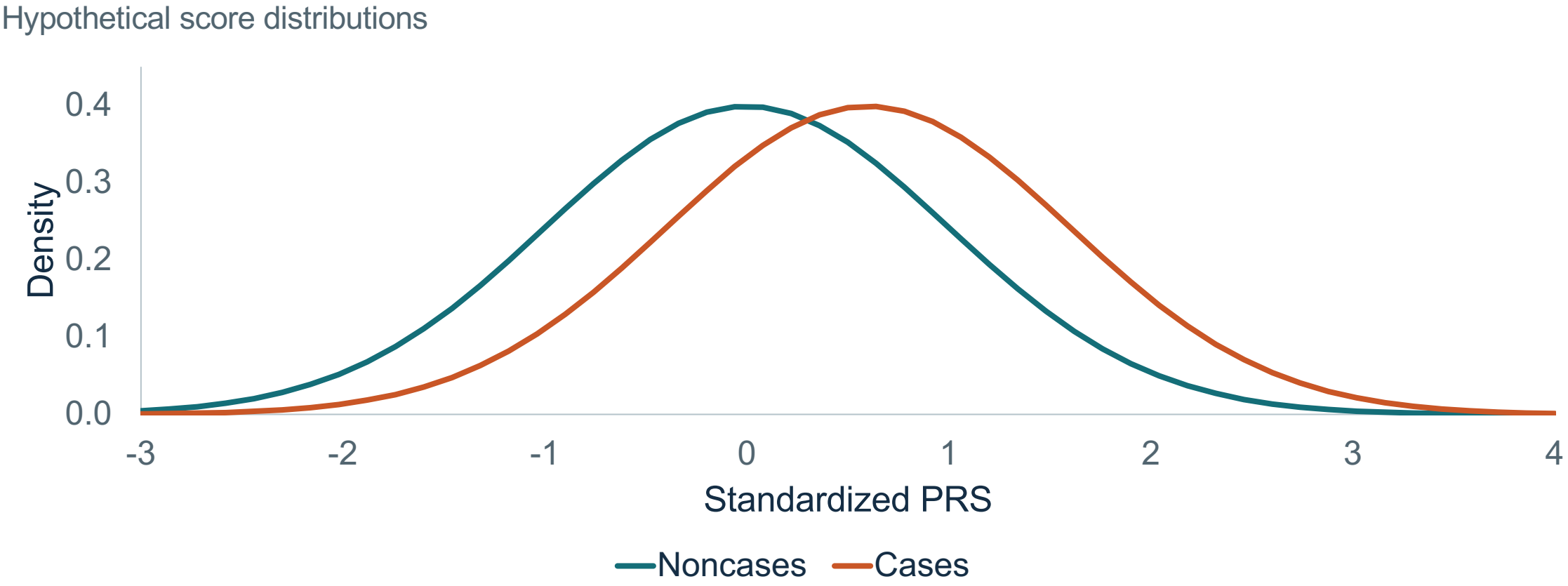
Four questions about performance

Association	Does risk change with PRS?
Discrimination	Are cases ranked above noncases?
Calibration	Do predicted risks match observed risks?
Decision value	Does the model improve decisions?

Each question needs its own evidence.

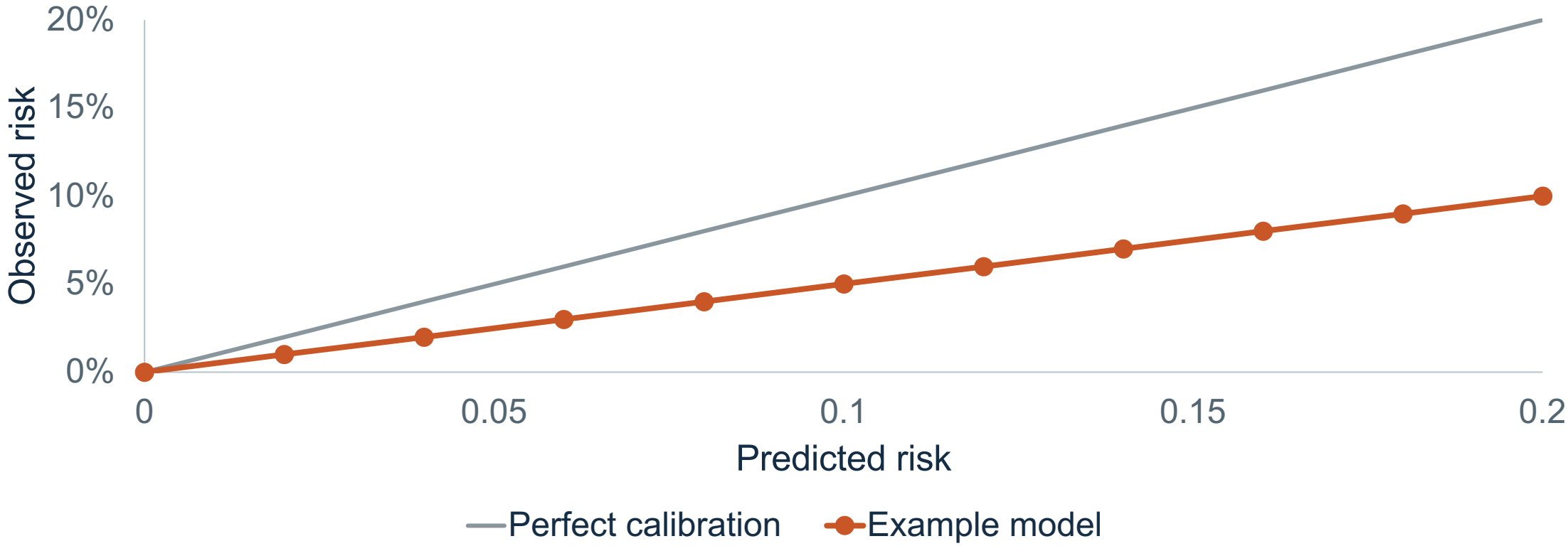
Wand et al., Nature, 2021; Vickers & Elkin, Med Decis Making, 2006.

Cases and noncases can have overlapping scores



AUC asks how often a randomly chosen case ranks above a noncase.

Good ranking can coexist with poor calibration

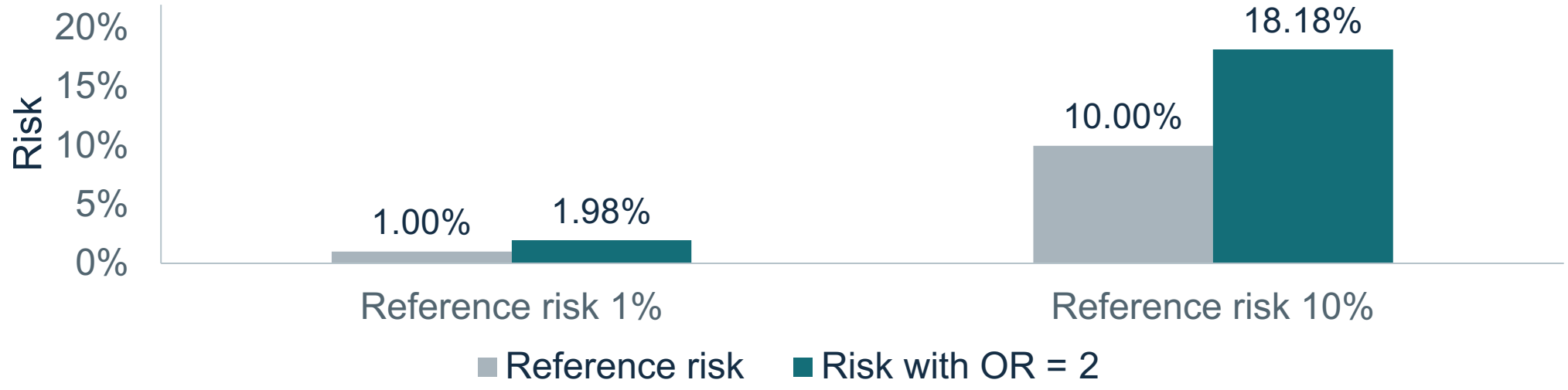


Hypothetical example: predicted risk 10%, observed risk 5%.

An odds ratio needs a reference risk

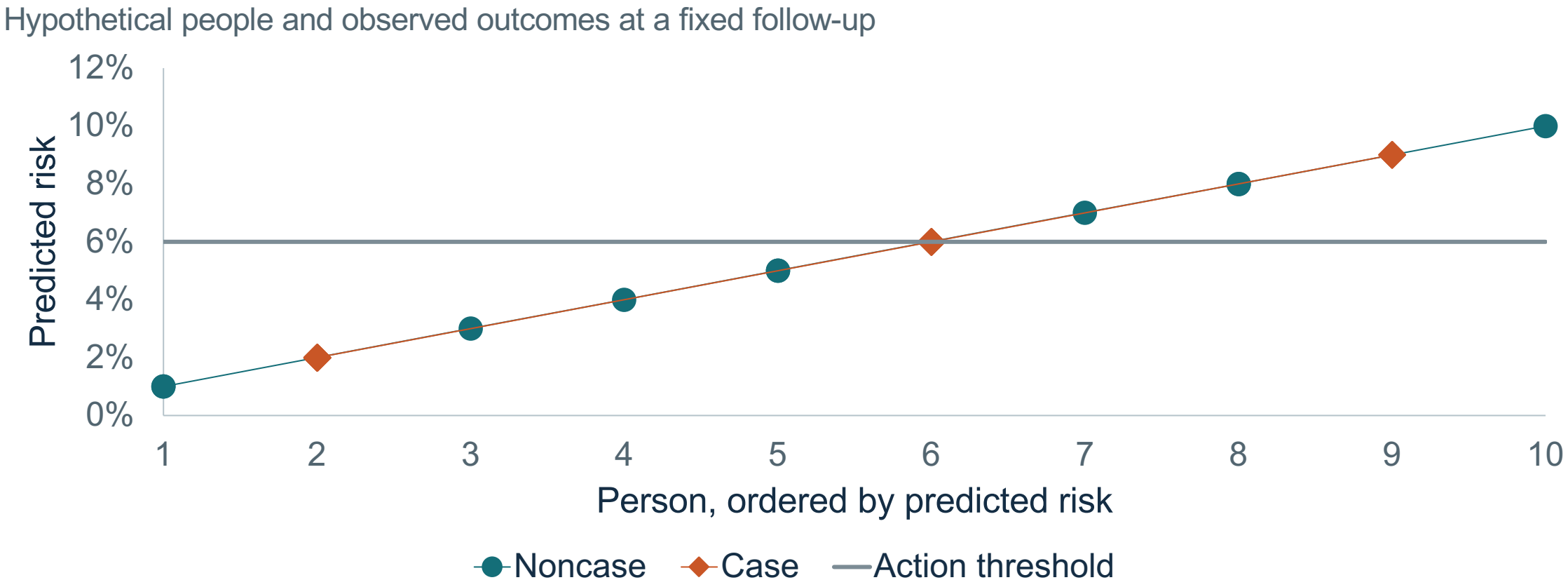
$$p_1 = \frac{\mathrm{OR} \times p_0}{1 - p_0 + \mathrm{OR} \times p_0}$$

p_0 : reference risk p_1 : resulting risk



OR = 2 gives 1.98% from 1%, but 18.18% from 10%.

A risk threshold creates a benefit-harm tradeoff



At ≥6%: 2 true positives, 3 false positives and 1 missed case.

Vickers & Elkin, Med Decis Making, 2006.

Activity: evaluate the claim

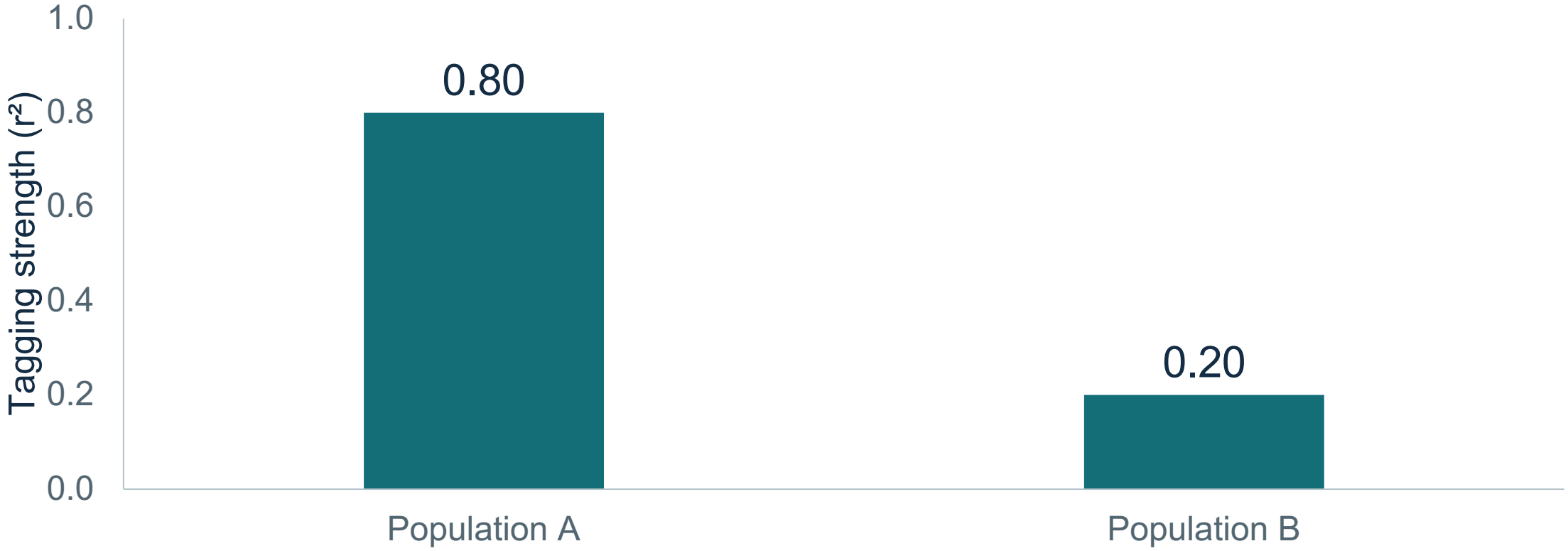
“AUC increased from 0.70 to 0.72, so the model is ready for screening.”

Which additional evidence would you request?

One minute in pairs. Separate prediction evidence from screening evidence.

The same causal effect can be tagged differently

Hypothetical tag-causal correlations. Causal effect held constant.



LD can change prediction even when the causal effect is unchanged.

Wang et al., Nat Commun, 2020.

Ancestry terminology

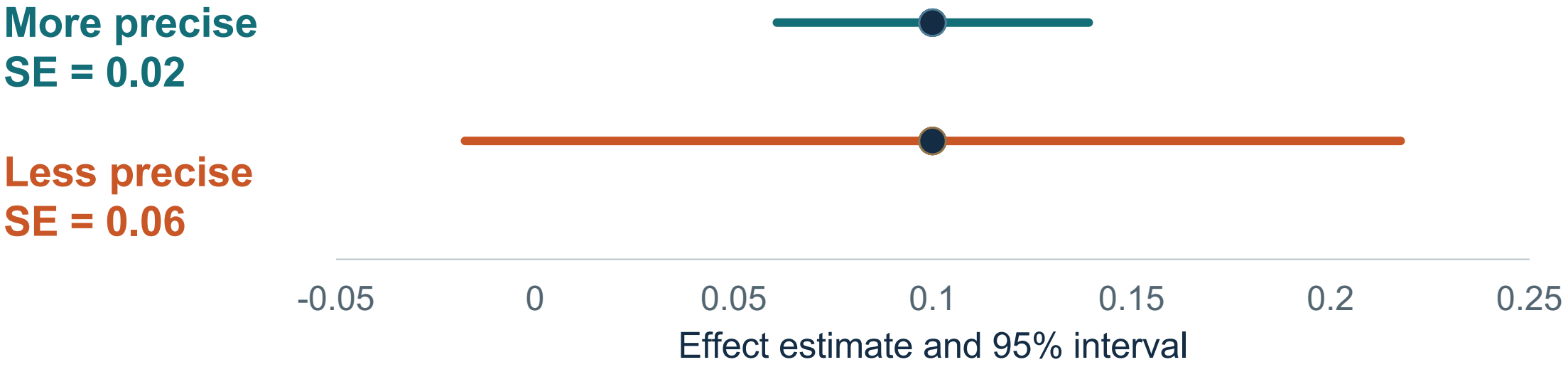
Genetic ancestry	Relationships in inherited variation
Principal components	Continuous coordinates of genetic similarity
Broad groups	Coarse summaries of a continuum
Race and ethnicity	Social and self-identified categories

A principal component is not an ancestry percentage.

Ding et al., Nature, 2023.

Larger GWAS can provide more precise information

Hypothetical estimates: 0.10, with intervals of ±1.96 SE



Multi-ancestry models can use shared evidence while allowing distinct effects.

Ruan et al., Nat Genet, 2022; Zhang et al., Nat Genet, 2023.

Population-specific effects and LD

$$\mathbf{b}_k = \mathbf{R}_k \boldsymbol{\beta}_k + \boldsymbol{\eta}_k, \qquad k = 1, \ldots, K$$

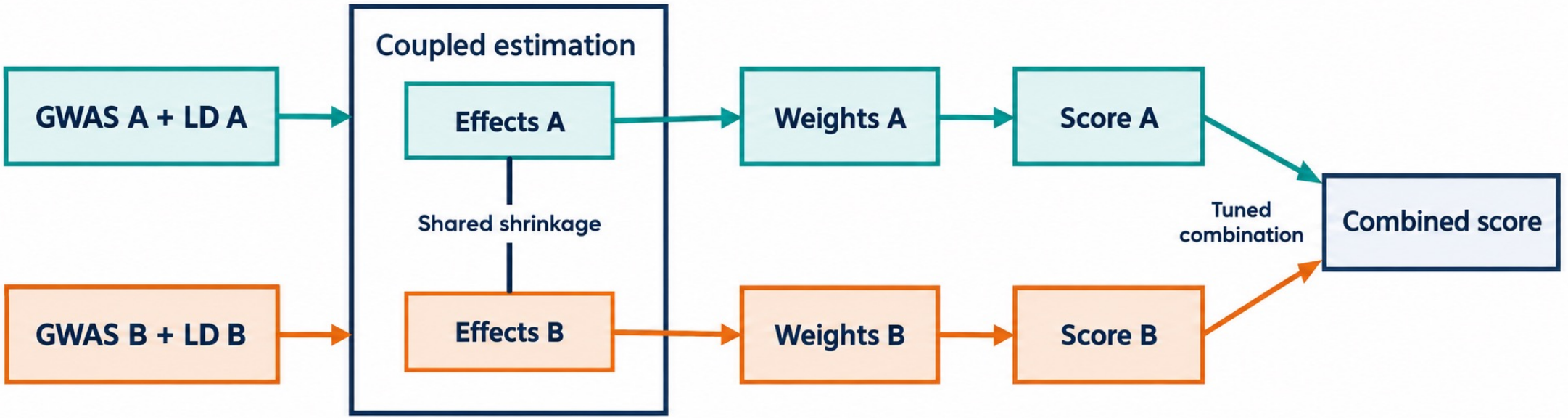
Each population k has its own effects β_k, LD matrix R_k and sample size N_k.

The methods differ in what they share across populations.

Sharing information can improve precision while retaining population-specific effects.

Ruan et al., Nat Genet, 2022; Zhang et al., Nat Genet, 2023; Zhang et al., Nat Commun, 2024.

PRS-CSx shares shrinkage across populations



Shared shrinkage, population-specific LD and effect estimates.

Ruan et al., Nat Genet, 2022.

PRS-CSx: a shared local shrinkage scale

$$\beta_{jk} \mid q_j, \sigma_k^2 \sim \mathcal{N}\left(0, \frac{\sigma_k^2}{N_k} \underbrace{q_j}_{\text{shared scale}}\right)$$

The local scale is shared. SNP effects and LD remain population-specific.

Evidence for a SNP changes its shared shrinkage scale.

Shared shrinkage does not require equal effects or matching effect directions.

Ruan et al., Nat Genet, 2022.

CT-SLEB: selection, empirical Bayes and ensemble

Clumping + thresholding

Select SNPs that pass a cutoff in either population.

Empirical Bayes

Borrow information through estimated effect covariance.

Super learning

Combine candidate scores in the tuning set.

z: GWAS z statistics. θ: underlying signals. Σ: cross-population covariance.

$$\widehat{\boldsymbol{\theta}}_j^{\mathrm{EB}} = \mathrm{E}\left(\boldsymbol{\theta}_j \mid \mathbf{z}_j; \widehat{\boldsymbol{\Sigma}}\right)$$

The method adjusts SNP weights, then combines candidate scores.

Zhang et al., Nat Genet, 2023.

PROSPER: shrink effects and their differences

$$\min_{\boldsymbol{\beta}_1,\boldsymbol{\beta}_2} \quad \underbrace{L_1(\boldsymbol{\beta}_1) + L_2(\boldsymbol{\beta}_2)}_{\text{fit GWAS and LD}} + 2\lambda_1\|\boldsymbol{\beta}_1\|_1 + 2\lambda_2\|\boldsymbol{\beta}_2\|_1 + c\|\boldsymbol{\beta}_1^S - \boldsymbol{\beta}_2^S\|_2^2$$

λ_1, λ_2: shrink SNP effects within each population.

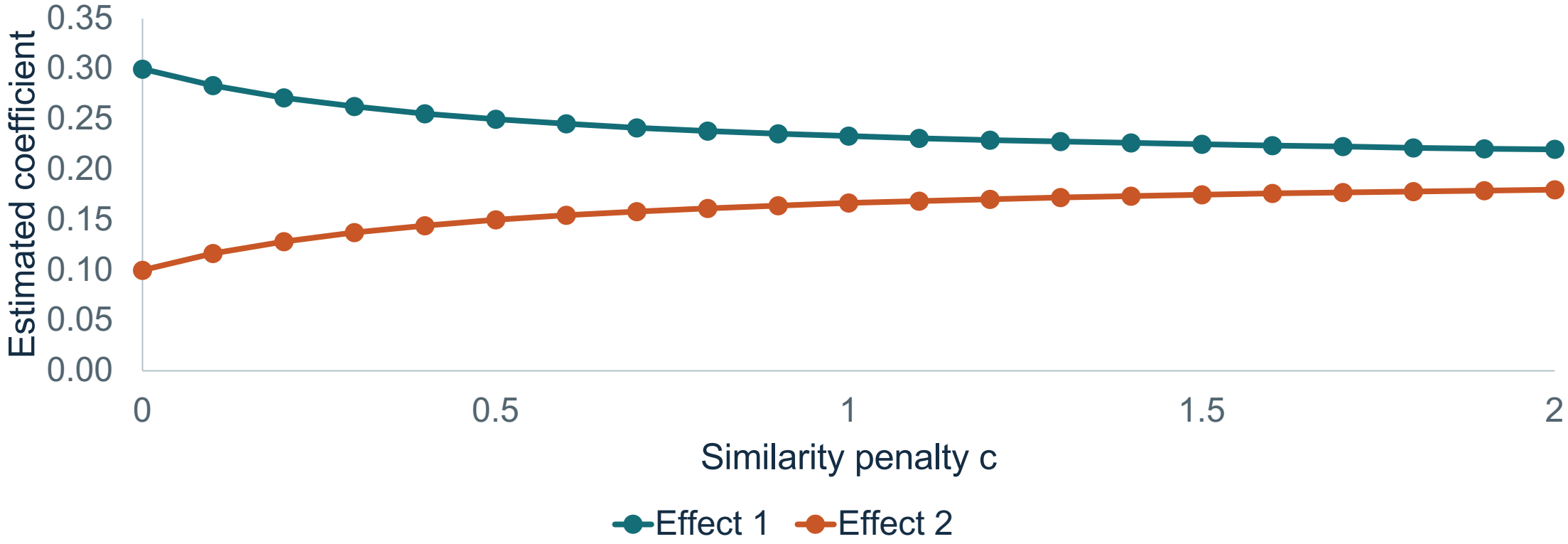
c: shrink differences between effects at shared SNPs S.

Single-ancestry fits set penalty scales. A final ensemble combines candidate scores.

Zhang et al., Nat Commun, 2024.

How a similarity penalty changes two estimates

One SNP, initial estimates 0.30 and 0.10. Larger c brings them closer.



Hypothetical illustration of one penalty term, with lasso and LD omitted.

Zhang et al., Nat Commun, 2024.

Three ways to share information

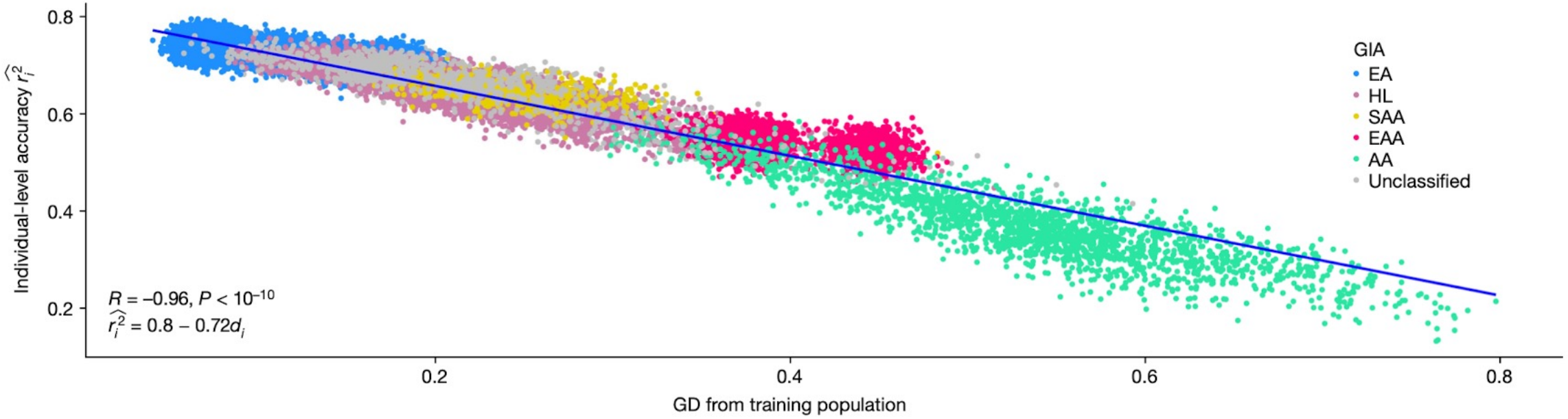
Method	What is shared?	How is LD used?
PRS-CSx	SNP-specific shrinkage scales	Joint model, population-specific LD
CT-SLEB	Cross-population effect covariance	Clumping before empirical Bayes
PROSPER	Penalized similarity of SNP effects	Population-specific quadratic loss

Compare frozen models in the same independent validation design.

Ruan et al., Nat Genet, 2022; Zhang et al., Nat Genet, 2023; Zhang et al., Nat Commun, 2024.

Prediction accuracy across the ancestry continuum

Height in ATLAS, using a model trained in UK Biobank



Accuracy relative to genetic liability, estimated from posterior uncertainty

Prediction accuracy can vary within broad ancestry groups.

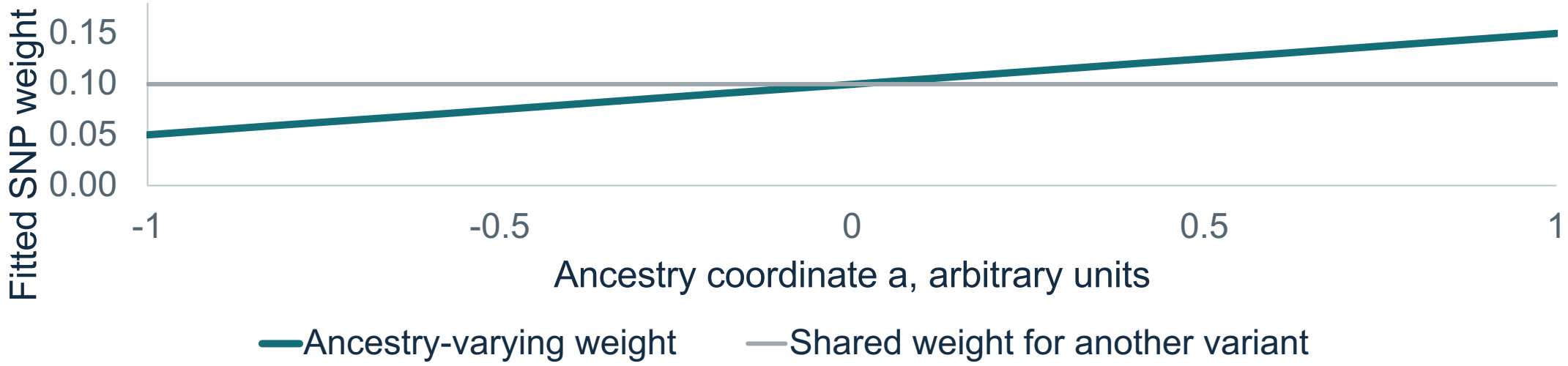
Ding et al., Nature, 2023, Fig. 3b.

SPLENDID: genotype-by-PC interactions

$$\mathrm{PRS}_i = \sum_{j=1}^{M} G_{ij} B_{ij}, \qquad B_{ij} = \beta_{j0} + \sum_{k=1}^{K} \beta_{jk} \mathrm{PC}_{ik}$$

Each SNP weight combines a shared effect and PC-dependent effects.

Hypothetical example, one SNP and one PC: $w(a) = 0.10 + 0.05a$



SNP groups are selected and shrunk, then candidate scores are combined.

Chen et al., Nat Methods, 2026. Hypothetical SNP example.

Rare variants can be grouped within genes

$$2Nf = 2 \times 100{,}000 \times 0.0001 = 20 \text{ expected allele copies}$$

N: diploid participants. f: allele frequency.

Four illustrative participants from a larger cohort

Person	Variant A	Variant B	Variant C	Burden
1	1	0	0	**1**
2	0	1	0	**1**
3	0	0	1	**1**
4	0	0	0	**0**

$$B_{ig} = \sum_{j \in S_g} G_{ij}$$

One gene and functional category define a qualifying set.

Grouping pools information across carriers of different variants.

Williams et al., Nat Commun, 2026. Illustrative allele counts.

RICE: common and rare score components

Component scores are estimated using discovery and tuning data.

RICE-CV

Ensemble existing common-variant scores

RICE-RV

Fit and ensemble gene/annotation burden scores

Common-variant score

Rare-variant score

Joint validation model

$$y_i = \alpha + b_C \, \mathrm{PRS}_{\mathrm{CV},i} + b_R \, \mathrm{PRS}_{\mathrm{RV},i} + \mathbf{z}_i^\top \boldsymbol{\delta} + \varepsilon_i$$

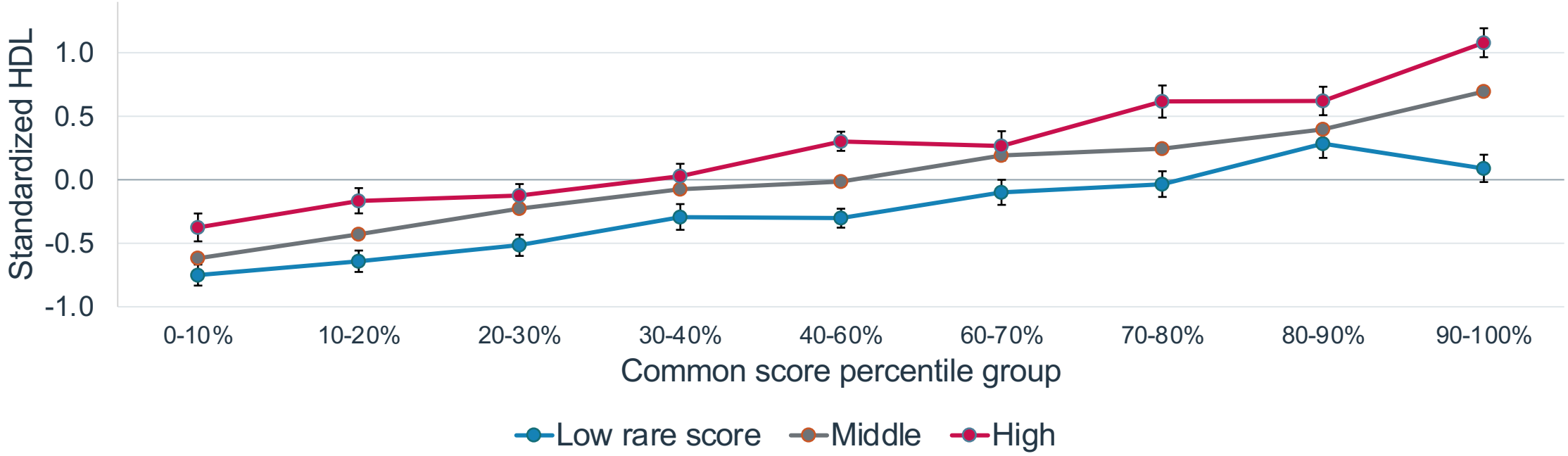
Component coefficients assess association conditional on the other score.

Williams et al., Nat Commun, 2026.

Rare scores further stratify common-score groups

HDL in UK Biobank validation, imputation plus exomes, N = 18,150

Rare-score groups: low <5th, middle 30th to 70th, high >95th percentile



Points show mean adjusted HDL. Error bars show ±1 standard error.

Rare scores distinguish HDL levels within common-score groups.

Williams et al., Nat Commun, 2026, Fig. 5a.

Five questions for a PRS study

Target	Which outcome, population and decision?
Independence	Was the final evaluation protected?
Reproducibility	Which alleles, weights and reference data?
Added value	Beyond the clinical model, with calibration?
Applicability	Does evidence support the proposed use?

PRS-RS and the PGS Catalog support transparent reporting.

Wand et al., Nature, 2021; Lambert et al., Nat Genet, 2021.

Discussion

Which PRS use matters in your research?

What baseline model and validation evidence would you require?

Four minutes for discussion. Suggested answers are in the instructor notes.

Backup: a simple shrinkage calculation

$$\mathrm{E}(\beta \mid \widehat{\beta}) = \frac{\tau^2}{\tau^2 + s^2}\,\widehat{\beta}$$

$\widehat{\beta}$	An approximately normal, unbiased effect estimate
s	Its standard error
τ^2	The prior variance of true effects

- More uncertainty in the estimate produces more shrinkage.
- This one-variant formula omits LD and other predictors.

Ge et al., Nat Commun, 2019.

Backup: PRS in a logistic risk model

$$\log\left(\frac{p}{1-p}\right) = \alpha + \gamma z + \text{clinical terms}$$

p	Outcome probability over the defined period
z	Standardized PRS using a stated reference distribution
$\alpha,\ \gamma$	Fitted intercept and score coefficient

- Clinical terms can include age and other prespecified predictors.
- For a one-SD score difference, the conditional odds ratio is exp(γ).

An intercept and a PRS coefficient must be calibrated for the population of use.

Backup: most cases may fall outside the highest decile

Group	People	Hypothetical risk	Expected cases
Highest PRS decile	1,000	4%	40
Remaining population	9,000	1%	90
Total	10,000		130

The highest decile contains 10% of people and 40 / 130 = 30.8% of expected cases.

Backup: decision curve analysis

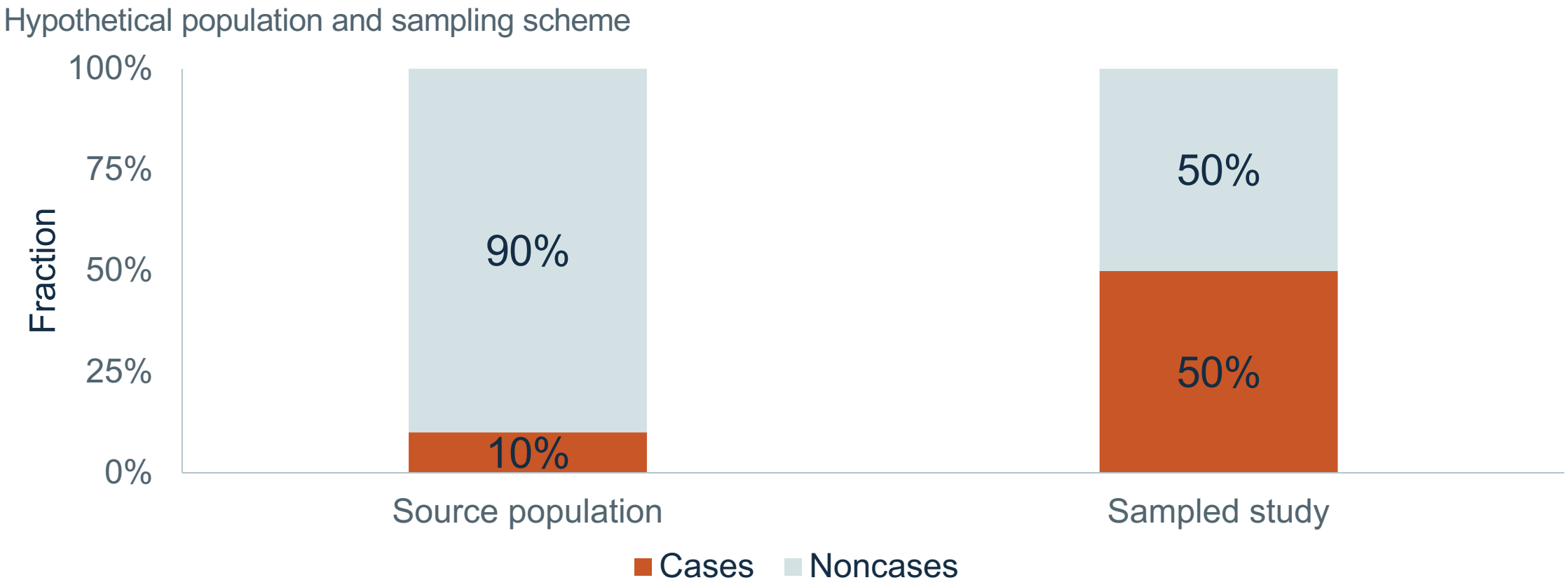
$$\text{Net benefit} = \frac{\text{TP}}{N} - \frac{\text{FP}}{N}\frac{t}{1-t}$$

$\text{TP},\ \text{FP}$	True positives and false positives under a decision rule
N	Number of people evaluated
t	Risk threshold at which an action is worthwhile

- Compare the model with relevant alternative strategies.
- Use clinically plausible thresholds and account for sampling.

Vickers & Elkin, Med Decis Making, 2006.

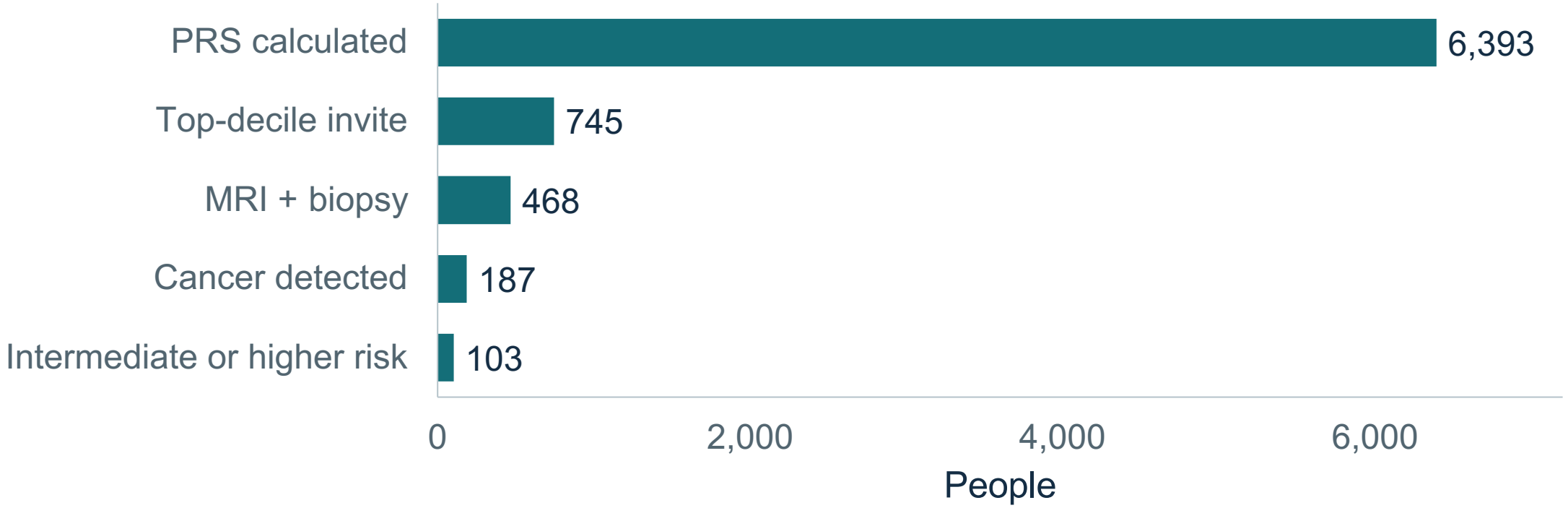
Backup: case-control sampling



The sampled case fraction does not give population disease risk.

Backup: BARCODE1: PRS-guided prostate screening

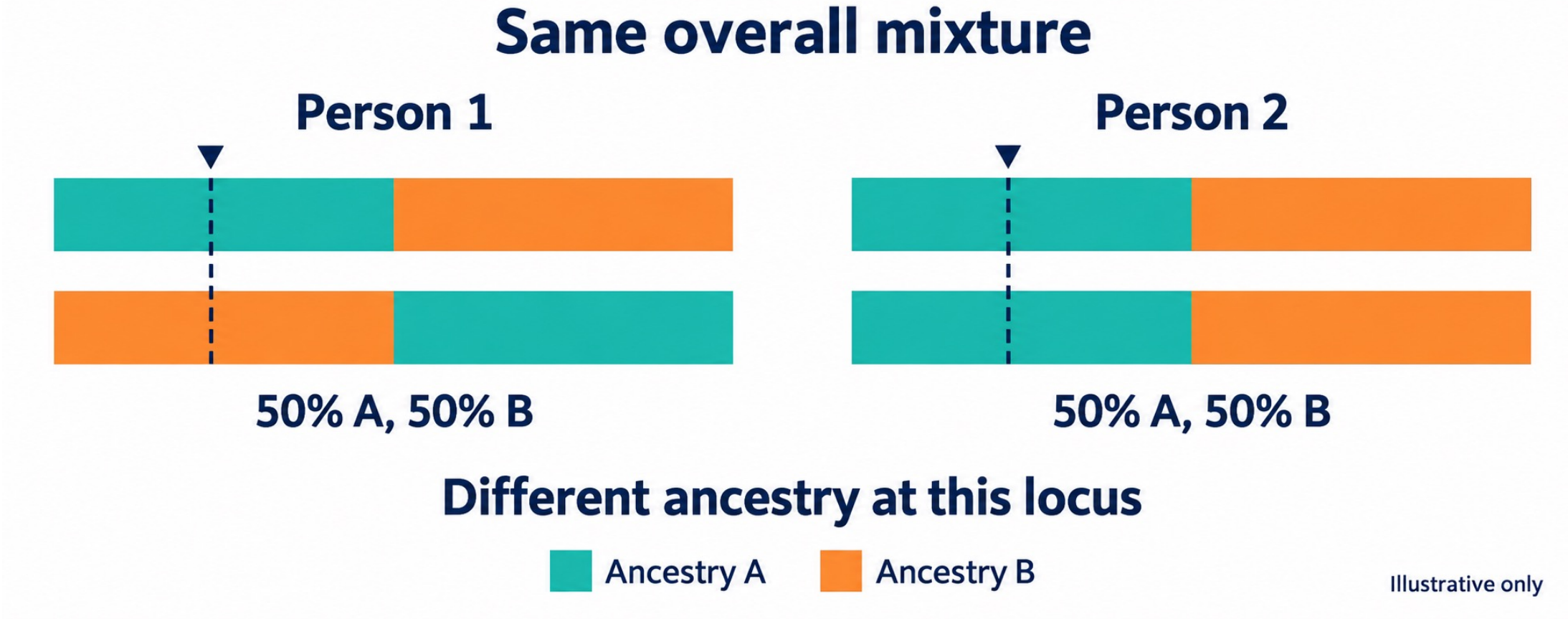
40,292 invited initially. Later stages are nested subsets.



Prospective single-arm study. Detection yield does not establish mortality benefit.

McHugh et al., N Engl J Med, 2025.

Backup: Global and local ancestry describe different information



GAUDI uses local ancestry. SPLENDID uses ancestry PCs.

Chen et al., Nat Methods, 2026; Sun et al., Nat Commun, 2024.

Backup: Who is hidden by an overall performance estimate?

For whom does the model work?

Report sample sizes, events and uncertainty across the relevant ancestry range.

Check calibration and absolute risk in the population of use.

Wand et al., Nature, 2021; Ding et al., Nature, 2023.

Backup: SPLENDID fitting and ensembling

Each SNP group contains its main effect and eligible PC interactions.

$$\min_{\boldsymbol{\beta},\boldsymbol{\alpha}} \; \mathcal{L}(\boldsymbol{\beta},\boldsymbol{\alpha}) + \underbrace{\lambda_0 \sum_{j=1}^{M} \mathbf{1}\{\boldsymbol{\beta}_j \neq 0\}}_{\text{select SNP groups}} + \underbrace{\lambda_1 \sum_{j=1}^{M} \|\boldsymbol{\beta}_j\|_2}_{\text{shrink SNP groups}}$$

Squared-error loss. Baseline covariates z include PC main effects.

$$\mathcal{L} = \frac{1}{2N} \sum_{i=1}^{N} \left(y_i - \mathbf{z}_i^\top \boldsymbol{\alpha} - \sum_{j=1}^{M} G_{ij} B_{ij} \right)^2$$

$$\mathrm{PRS}_i^{\mathrm{ensemble}} = \sum_{d=1}^{D} a_d \, \mathrm{PRS}_i^{(d)}$$

Candidate fits vary in penalty settings. Tuning learns ensemble weights.

Chen et al., Nat Methods, 2026. Adapted from Chen dissertation defense, slides 23 and 25.

Backup: The evaluation depends on who carries the variants

Population gain and carrier gain can differ

Report both the overall improvement and performance in relevant carriers.

Small carrier counts can produce substantial uncertainty.

Williams et al., Nat Commun, 2026; Nam et al., Nat Genet, 2026.

Backup: single-ancestry methods

Method	How it handles the signal	Key inputs or choices
Clumping + thresholding	Selects a relatively independent subset	LD settings and p-value cutoff
LDpred2	Models LD with a mixture prior on effects	GWAS, LD, genetic architecture parameters
PRS-CS	Models LD with adaptive continuous shrinkage	GWAS, LD, shrinkage setting or auto estimation

Method choice should follow data quality and independent comparison.

Ge et al., Nat Commun, 2019; Privé et al., Bioinformatics, 2020.

Backup: Combining two scores for one person

$$\mathrm{PRS}_{\mathrm{combined}} = w_1 S_1 + w_2 S_2$$

Component	Score	Fitted coefficient	Contribution
Score 1	0.80	0.70	0.56
Score 2	0.30	0.50	0.15

Combined score = 0.71. These coefficients are not ancestry proportions.

Ruan et al., Nat Genet, 2022; Zhang et al., Nat Genet, 2023; Hoggart et al., Nat Genet, 2024.

Backup: summary-statistic likelihood

$$\boldsymbol{\eta} = \frac{\mathbf{X}^\top \boldsymbol{\epsilon}}{N}, \qquad \mathrm{Cov}(\boldsymbol{\epsilon} \mid \mathbf{X}) = \sigma^2 \mathbf{I}_N$$

$$\mathrm{Cov}(\boldsymbol{\eta} \mid \mathbf{X}) = \frac{\sigma^2 \mathbf{X}^\top \mathbf{X}}{N^2} = \frac{\sigma^2}{N}\mathbf{R}$$

$$\mathbf{b} \mid \boldsymbol{\beta}, \mathbf{X} \sim \mathcal{N}\left(\mathbf{R}\boldsymbol{\beta}, \frac{\sigma^2}{N}\mathbf{R}\right)$$

N: participants. M: variants. R is M × M.

The same LD matrix appears in the mean and the sampling covariance.

This is the standardized quantitative-trait working model.

Ge et al., Nat Commun, 2019; Privé et al., Bioinformatics, 2020.

Backup: PRS-CS global-local hierarchy

$$\beta_j \mid \phi, \psi_j, \sigma^2 \sim \mathcal{N}\left(0, \frac{\sigma^2}{N}\phi\psi_j\right)$$

$$\psi_j \mid \delta_j \sim \text{Gamma}(a, \delta_j), \quad \delta_j \sim \text{Gamma}(b, 1)$$

$$\text{E}(\boldsymbol{\beta} \mid \phi, \mathbf{D}, \mathbf{b}) = \left[\mathbf{R} + (\phi\mathbf{D})^{-1}\right]^{-1}\mathbf{b}$$

Gamma distributions use shape and rate. PRS-CS uses a = 1 and b = 1/2.

D contains the local scales. The full algorithm averages over their uncertainty.

This conditional mean explains the mechanism, not the full sampler.

Ge et al., Nat Commun, 2019.

Backup: PRS-CSx shared-scale hierarchy

$$\beta_{jk} \mid q_j, \sigma_k^2 \sim \mathcal{N}\left(0, \frac{\sigma_k^2}{N_k} q_j\right)$$

$$q_j \mid \delta_j \sim \text{Gamma}(1, \delta_j)$$

$$\delta_j \sim \text{Gamma}\left(\frac{1}{2}, \phi\right)$$

Each SNP has one scale shared across populations indexed by k.

The global parameter φ enters the scale hierarchy.

Do not multiply by φ again in the normal variance under this parameterization.

Ruan et al., Nat Genet, 2022.

Backup: CT-SLEB empirical Bayes step

$$z_{jk} = \frac{\widehat{\gamma}_{jk}}{\mathrm{SE}_{\gamma,jk}}, \qquad \mathbf{z}_j \mid \boldsymbol{\theta}_j \sim \mathcal{N}(\boldsymbol{\theta}_j, \mathbf{I}_K)$$

$$\widehat{\boldsymbol{\theta}}_j^{\mathrm{EB}} = \widehat{\boldsymbol{\Sigma}}\left(\widehat{\boldsymbol{\Sigma}} + \mathbf{I}_K\right)^{-1}\mathbf{z}_j$$

$$\widehat{\gamma}_{jk}^{\mathrm{EB}} = \mathrm{SE}_{\gamma,jk}\ \widehat{\theta}_{jk}^{\mathrm{EB}}$$

Gamma-hat is the raw per-allele estimate, with its matching standard error.

Σ describes K × K signal covariance. Independent sampling noise has covariance I.

The target component is transformed back to the dosage scale before scoring.

Zhang et al., Nat Genet, 2023.

Backup: PROSPER summary-data objective

$$L_k(\boldsymbol{\beta}_k) = \boldsymbol{\beta}_k^\top (\mathbf{R}_k + \delta_k \mathbf{I}) \boldsymbol{\beta}_k - 2\mathbf{b}_k^\top \boldsymbol{\beta}_k$$

$$Q(\boldsymbol{\beta}_1, \boldsymbol{\beta}_2) = \sum_{k=1}^{2} [L_k(\boldsymbol{\beta}_k) + 2\lambda_k \|\boldsymbol{\beta}_k\|_1] + c\|\boldsymbol{\beta}_1^S - \boldsymbol{\beta}_2^S\|_2^2$$

δ_k regularizes the LD matrix. S contains aligned SNPs shared by both populations.

Single-ancestry fits define relative penalties. Tuning fits the final ensemble.

The method estimates coefficients by optimization, then combines candidate scores.

Zhang et al., Nat Commun, 2024.

Backup: additional directions in PRS

Direction	What it adds	Example or consideration
Functional information	Annotations inform variant prioritization or shrinkage	Requires independent evaluation
Multiple traits	Borrows information from correlated phenotypes	Check overlap and phenotype relevance
Ancestry interpolation	Interpolates fine-tuned scores in ancestry space	DiscoDivas (2026)
Rare-variant modeling	Learns annotation-based scores or shrinks rare effects	DeepRVAT (2024), RareEffect (2026)

These extensions use the same principles of estimation, validation and transportability.

Ruan et al., Am J Hum Genet, 2026; Clarke et al., Nat Genet, 2024; Nam et al., Nat Genet, 2026.

Backup: a PGS Catalog reading exercise

01	**Find**	Open PGS000004, the PRS313 breast cancer score.
02	**Describe**	Identify the phenotype, variant count and development samples.
03	**Inspect**	Find the scoring-file allele and weight fields, and genome build.
04	**Appraise**	Find independent evaluations relevant to a proposed population.

Resource: pgscatalog.org/score/PGS000004/

Lambert et al., Nat Genet, 2021; Mavaddat et al., Am J Hum Genet, 2019.

Reading: single-ancestry methods

Choi et al., Nat Protoc, 2020. Tutorial: a guide to performing polygenic risk score analyses

https://doi.org/10.1038/s41596-020-0353-1

Ge et al., Nat Commun, 2019. Polygenic prediction via Bayesian regression and continuous shrinkage priors

https://doi.org/10.1038/s41467-019-09718-5

Privé et al., Bioinformatics, 2020. LDpred2: better, faster, stronger

https://doi.org/10.1093/bioinformatics/btaa1029

Reading: multi-ancestry methods

Ruan et al., Nat Genet, 2022. Improving polygenic prediction in ancestrally diverse populations

https://doi.org/10.1038/s41588-022-01054-7

Zhang et al., Nat Genet, 2023. A new method for multiancestry polygenic prediction improves performance across diverse populations

https://doi.org/10.1038/s41588-023-01501-z

Zhang et al., Nat Commun, 2024. An ensemble penalized regression method for multi-ancestry polygenic risk prediction

https://doi.org/10.1038/s41467-024-47357-7

Hoggart et al., Nat Genet, 2024. BridgePRS leverages shared genetic effects across ancestries to increase polygenic risk score portability

https://doi.org/10.1038/s41588-023-01583-9

Reading: ancestry and continuous models

Wang et al., Nat Commun, 2020. Theoretical and empirical quantification of the accuracy of polygenic scores in ancestry divergent populations

https://doi.org/10.1038/s41467-020-17719-y

Ding et al., Nature, 2023. Polygenic scoring accuracy varies across the genetic ancestry continuum

https://doi.org/10.1038/s41586-023-06079-4

Chen et al., Nat Methods, 2026. SPLENDID incorporates continuous genetic ancestry in biobank-scale data to improve polygenic risk prediction across diverse populations

https://doi.org/10.1038/s41592-026-03235-2

Sun et al., Nat Commun, 2024. Improving polygenic risk prediction in admixed populations by explicitly modeling ancestral-differential effects via GAUDI

https://doi.org/10.1038/s41467-024-45135-z

Ruan et al., Am J Hum Genet, 2026. DiscoDivas: Leveraging genetic-ancestry continuum information to interpolate PRS for admixed populations

https://doi.org/10.1016/j.ajhg.2026.05.006

Reading: rare variants and risk prediction

Williams et al., Nat Commun, 2026. Integrating common and rare variants improves polygenic risk prediction across diverse populations

https://doi.org/10.1038/s41467-026-72185-2

Clarke et al., Nat Genet, 2024. Integration of variant annotations using deep set networks boosts rare variant association testing

https://doi.org/10.1038/s41588-024-01919-z

Nam et al., Nat Genet, 2026. Rare variant effect estimation and polygenic risk prediction

https://doi.org/10.1038/s41588-026-02705-9

Fahed et al., Nat Commun, 2020. Polygenic background modifies penetrance of monogenic variants for tier 1 genomic conditions

https://doi.org/10.1038/s41467-020-17374-3

McHugh et al., N Engl J Med, 2025. Assessment of a Polygenic Risk Score in Screening for Prostate Cancer

https://doi.org/10.1056/NEJMoa2407934

Reading: reporting and risk interpretation

Wand et al., Nature, 2021. Improving reporting standards for polygenic scores in risk prediction studies

https://doi.org/10.1038/s41586-021-03243-6

Lambert et al., Nat Genet, 2021. The Polygenic Score Catalog as an open database for reproducibility and systematic evaluation

https://doi.org/10.1038/s41588-021-00783-5

Vickers & Elkin, Med Decis Making, 2006. Decision curve analysis: a novel method for evaluating prediction models

https://doi.org/10.1177/0272989X06295361

Mavaddat et al., Am J Hum Genet, 2019. Polygenic Risk Scores for Prediction of Breast Cancer and Breast Cancer Subtypes

https://doi.org/10.1016/j.ajhg.2018.11.002